Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/27/2017
PI Name:	Williams, Forman A. Ph.D.		
Project Title:	Flammability Limit Experiment (FLEX) - Phase II		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	fwilliams@eng.ucsd.edu	**Fax:**	FY 858 534-5354
PI Organization Type:	UNIVERSITY	**Phone:**	858 534-5492
Organization Name:	University of California, San Diego		
PI Address 1:	Center for Energy Research		
PI Address 2:	Department of Mechanical and Aerospace Engineering		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92093-0411	**Congressional District:**	53
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	08/12/2016	**End Date:**	09/30/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hicks, Michael C.	**Contact Phone:**	216-433-6576
Contact Email:	Michael.C.Hicks@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2019 per NSSC information (Ed., 5/14/18) NOTE: End date changed to 1/31/2018 per M. Hicks/GRC and NSSC information (Ed., 9/26/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dryer, Frederick Ph.D. (Experiment Co-I: Princeton University) Choi, Mun Young Ph.D. (Experiment Co-I: University of Connecticut) Shaw, Benjamin Ph.D. (Experiment Co-I: University of California, Davis) Avedisian, C. Thomas Ph.D. (Experiment Co-I: Cornell University) Farouk, Tanvir Ph.D. (Experiment Co-I: University of South Carolina) Hicks, Michael M.S. (Experiment Co-I: NASA Glenn Research Center) Dietrich, Daniel Ph.D. (Experiment Co-I: NASA Glenn Research Center) Nayagam, Vedha Ph.D. (Experiment Co-I: Case Western Reserve University)		
Grant/Contract No.:	NNX16AR09A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note October 2016: Continues "Flammability Limit Experiment (FLEX)," grant NNX09AM33A] The overall objective of the FLEX experiments (Flammability Limit Experiment) is to improve fundamental knowledge of droplet burning and other combustion processes relevant to fire safety in manned space environments. Specific objectives are to investigate influences of low-gravity environments on flammability conditions of different fuels in different oxidizing atmospheres representative of fire-suppression scenarios. The first phase of FLEX was a focused set of science objectives designed to address spacecraft fire safety concerns. The primary objectives were the assessment of flammability limits in a range of habitable atmospheres and fire suppressant efficacies. The second phase is FLEX-2 with a more expansive set of science objectives designed to investigate fundamental phenomena associated with (i) fuel sooting, (ii) development of reference fuels for fuel surrogates, (iii) low flow convective effects on diffusion flames, (iv) spacecraft fire safety, (v) droplet interactions in spray combustion, and (vi) condensed phase effects on burning using bi-component fuel blends. The research team for FLEX comprises externally funded Principal Investigators (PIs) and internally funded PIs. The lead researcher is Forman Williams for both FLEX and FLEX-2 and can be considered the de facto “PI” for FLEX (i.e., the combined experiments). The externally funded PIs are grantees affiliated with universities. The internally funded PIs are Michael C. Hicks (NASA-Glenn Research Center-GRC), Daniel L. Dietrich (NASA-GRC), and Vedha Nayagam (Case Western Reserve University). Flammability Limit Experiment (FLEX) Experiment PI: Forman Williams, Ph.D., University of California San Diego. FLEX and FLEX-2 Experiment External CoInvestigators (CoIs): Frederick Dryer, Ph.D., Princeton University. Task title: "Scientific Support for a Space Based Droplet Burning Experiment" Mun Young Choi, Ph.D., University of Connecticut. Task title: "FLEX Droplet Combustion Experiments in Microgravity" FLEX-2 Experiment External CoInvestigators (CoIs): C. Thomas Avedisian, Ph.D., Cornell University. Task title: "Combustion of Surrogate Fuel Droplets in Microgravity" Benjamin Shaw, Ph.D., University of California Davis. Task title: "Scientific Support for the FLEX-2 ISS (International Space Station) Experiment" Tanvir Farouk, Ph.D., University of South Carolina. Task title: "Mathematical Modeling and Simulation of Space ased Droplet Combustion – Scientific Support to FLEX and FLEX-2 Program" FLEX and FLEX-2 Experiment Internal CoInvestigators (CoIs): Michael C. Hicks, MSME, NASA Glenn Research Center. Daniel L. Dietrich, Ph.D., NASA Glenn Research Center. Vedha Nayagam, Ph.D., Case Western Reserve University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth by improving knowledge of combustion processes needed by humankind.
Task Progress:	The overall objective of the FLEX experiments (Flammability Limit Experiment) is to improve fundamental knowledge of droplet burning and other combustion processes relevant to fire safety in manned space environments. Specific objectives are to investigate influences of low-gravity environments on flammability conditions of different fuels in different oxidizing atmospheres representative of fire-suppression scenarios. Experiments with hepane as the fuel and with methanol as the fuel are in progress aboard the ISS. During the reporting period, there was extensive discussion about the specific tests that were performed on ISS. The chemistry of combustion of the fuel affects the combustion and extinction conditions. Research at University of California, San Diego (UCSD) addressed problems of improving chemical-kinetic descriptions. In particular, extensive consideration of cool-flame chemistry in normal alkane droplet combustion was completed. This aspect of the research will be a major activity in future work. The project, in general, involves basic flammability-limit investigations relevant to fire-safety issues for manned space flight. In standard fire-suppression techniques, a suppressant jet is directed towards a fire or a smoldering region. In considering these techniques for space applications, it is necessary to analyze the effect of suppressant transport to both the near-field and far-field of the region of interest. Fire prevention can also be implemented by modifying the atmospheric composition of the environment so as to reduce the likelihood of a self-sustaining combustion event. Finally, the flammability properties of materials can also be modified by the addition of materials which suppress ignitability, enhance charring of surfaces, or chemically inhibit self-sustained burning. The research addressed these aspects of fire safety in space.
Bibliography Type:	Description: (Last Updated: 03/07/2024)
Articles in Peer-reviewed Journals	Cuoci A, Saufi AE, Frassoldati A, Dietrich DL, Williams FA, Faravelli T. "Flame extinction and low-temperature combustion of isolated fuel droplets of n-alkanes." Proceedings of the Combustion Institute. 2017;36(2):2531-9. https://doi.org/10.1016/j.proci.2016.08.019 , Jan-2017
Articles in Peer-reviewed Journals	Dietrich DL, Calabria R, Massoli P, Nayagam V, Williams FA. "Experimental observations of the low temperature burning of decane/hexanol droplets in microgravity." Combustion Science and Technology. 2017;189(3):520-54. http://dx.doi.org/10.1080/00102202.2016.1225730 , Jan-2017

Articles in Peer-reviewed Journals	Seshadri K, Peters N, Williams FA, Nayagam V, Paczko G. "Asymptotic analysis of quasi-steady n-heptane droplet combustion supported by cool-flame chemistry." Combustion Theory and Modelling. 2016;20(6):1118-30. Published online: 11 Nov 2016. http://dx.doi.org/10.1080/13647830.2016.1248494 , Nov-2016